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Depression screening with PHQ-9 combined with facial emotion analysis

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Abstract

Depression is one of the psychiatric diseases that has a high impact on the world population, including the Thai population. Suicide statistics around the world show a continuous increasing trend, with more than 700,000 people worldwide completing suicide. Therefore, screening to detect depression and provide treatment is crucial. Currently, the Ministry of Public Health and the Department of Mental Health in Thailand use the Patient Health Questionnaire-9 (PHQ-9) as a standard tool for screening and assessing the severity of such symptoms [1]. This research presents the development of a tool for screening and assessing the severity of depressive symptoms. We introduce the development of a web application to make screening for depressed patients efficient and easily accessible. Each PHQ-9 question is presented in a story format, and answers are selected based on the situations presented. Additionally, facial emotions are assessed throughout the screening period via the device's camera. We apply image processing technology (Image Processing) to detect movements of facial expression muscles, making screening and assessing the severity of depressive symptoms more effective. The sample group for testing was selected from individuals aged 15-29 years, as this group has the highest number of suicides among people with depression, totaling 52 individuals. It was found that depression screening with PHQ-9 combined with facial emotion analysis can help individuals feel more relaxed. Statistical data values for the results of the satisfaction evaluation were in the high range (average 4.10 and standard division 0.44).

Keywords: Depression, screen, facial emotions.

1. Introduction

The incidence of mental health problems and violence in society is increasing. Acts of violence often manifest verbally, causing psychological effects and physical harm to oneself and others. This issue of violence impacts individuals in families, communities, and society at large. The World Health Organization (WHO) is addressing this growing concern of violence, recognizing it as a leading cause of death and injury among individuals, with mortality rates on the rise.

Depression is one of the psychiatric diseases that has a high impact on the world population, including the Thai population. Suicide statistics around the world show a continuous increasing trend, with more than 700,000 people worldwide completing suicide. On average, one person commits suicide every 40 seconds.

In Thailand, as of 2021, 1.5 million people aged 15 and older experienced psychotic-depressive symptoms, leading to over 53,000 suicide attempts attributed to depression and approximately 4,000 resultant deaths. The most recent data from 2022 reveals that over 50,000 individuals contemplate suicide annually, with around 4,800 completing suicide each year. Many of these challenges originate from depression [2]. Individuals with depression have a suicide rate 20 times higher than the general population, and 70 percent of those with depression die prematurely. Many individuals with depression are unaware of their condition, thus not receiving proper treatment. Severe cases can lead to suicide attempts, corroborating WHO data. It is projected that by 2029, more than 5% of the general population will experience depression. Depression is anticipated to become the leading cause of disease burden globally and the second leading cause of suicide among individuals aged 15-29, constituting the youth demographic, a crucial force in a nation's development [3].

Screening and assessing the severity of primary depression symptoms have been limited to a self-report assessment of 9 questions (PHQ-9) by the Ministry of Public Health and the Department of Mental Health. Patients may minimize or exaggerate their answers, which may not accurately reflect their true feelings. To improve the efficiency of assessment results and make the assessment form more accessible, this research introduces a screening tool for patients with depression. The tool is developed in

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the form of a web application, where each question of the PHQ-9 assessment is presented through a story with accompanying text, images, and instrumental music. During evaluation, facial expressions are assessed via a camera on a computer or mobile device. Subsequently, the score from each question in the PHQ-9 assessment is considered alongside the results of emotional state processing throughout the assessment period.

2. Theory and research related

Depression is a serious mood disorder that affects individuals' thoughts, feelings, and behavior. Depressive disorder is characterized by persistent feelings of sadness, hopelessness, and disinterest in activities. This condition can have significant psychosomatic symptoms and can impair an individual's ability to function in daily life [4].

Factors that cause depression in Thai people include four factors [5] as follows:

1. Personal factors, which include gender, age, marital status, religion, occupation or work history, substance use, financial problems or income adequacy, and history of physical or mental health problems.
2. Biological factors, such as family history of depression and symptoms during menopause.
3. Psychosocial factors, including loss of a mother before age 11, negative automatic thoughts, feelings of guilt, management of internal stimuli, and sense of belonging.
4. Social factors, such as stressful life events, perception of stress, interpersonal conflict, and social support.

The diagnosis or evaluation of primary depression symptoms by the Ministry of Public Health and the Department of Mental Health in Thailand uses a self-report assessment form consisting of 9 questions (PHQ-9) based on the patient's symptoms. Each question can be answered as follows: Not at all (0 points), Happens some days, not often (1 point), Quite often (2 points), and Every day (3 points).

The severity of depression symptoms is divided into four levels based on the total score: less than 7 points indicates no symptoms of depression, 7-12 points indicate a mild level, 13-18 points indicate a moderate level, and 19 points or more indicate symptoms of severe depression [6].

PATIENT HEALTH QUESTIONNAIRE-9 (PHQ-9)

Over the last 2 weeks, how often have you been bothered by any of the following problems? *(Use "✔" to indicate your answer)*	Not at all	Several days	More than half the days	Nearly every day
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, or hopeless	0	1	2	3
3. Trouble falling or staying asleep, or sleeping too much	0	1	2	3
4. Feeling tired or having little energy	0	1	2	3
5. Poor appetite or overeating	0	1	2	3
6. Feeling bad about yourself — or that you are a failure or have let yourself or your family down	0	1	2	3
7. Trouble concentrating on things, such as reading the newspaper or watching television	0	1	2	3
8. Moving or speaking so slowly that other people could have noticed? Or the opposite — being so fidgety or restless that you have been moving around a lot more than usual	0	1	2	3
9. Thoughts that you would be better off dead or of hurting yourself in some way	0	1	2	3

FOR OFFICE CODING __0__ + _____ + _____ + _____
=Total Score: _____

If you checked off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all	Somewhat difficult	Very difficult	Extremely difficult
☐	☐	☐	☐

Figure 1 the Patient Health Questionnaire-9 (PHQ-9)

Emotions and facial expressions [7]

The face is the body's primary medium for expressing emotions rapidly and effectively through the movement of its various elements, such as the eyes, eyelids, eyebrows, nose, and mouth. It stands as the most intricate part of the body's signaling system, composed of over 40 muscles that function independently of each other.

The main muscles used to make facial expressions include [8]:

- Orbicularis oculi: Surrounding the eye, it closes the eyelids and raises the eyebrows.
- Zygomaticus: Located in the cheek, it enables smiling and laughing.
- Levator labii: Situated in the upper lip, it raises the upper lip.
- Depressor labii: Found in the lower lip, it lowers the lower lip.
- Frontalis: Located in the forehead, it raises the eyebrows and creates wrinkles on the forehead.
- Corrugator supercilii: Also in the forehead, it creates wrinkles and furrows between the eyebrows.
- Orbicularis oris: Surrounding the mouth, it puckers and purses the lips.

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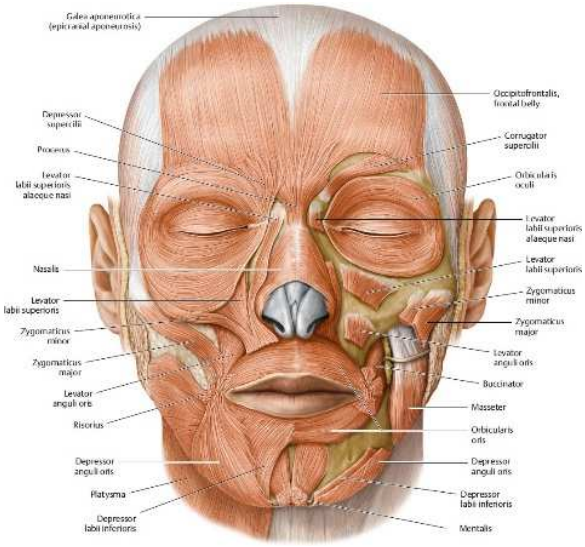

Figure 2 the muscles of facial expression.[8].

To address these limitations and enhance depression screening, researchers have explored alternative approaches. Sukontip K. et al. (2015) developed a face mood pictorial depression assessment tool to determine the sensitivity and specificity of a picture-based approach. Their study, involving 100 individuals receiving services at Suan Prung Hospital, utilized images representing various emotions to assess depression levels. The Picture Depression Assessment Toolkit demonstrated relatively high accuracy in screening for depression and was found to be less time-consuming than traditional assessments like the PHQ-9 [9]. In 2018, Sarira El-Den et al. conducted a systematic review titled "The psychometric properties of depression screening tools in primary healthcare settings," emphasizing the extensive psychometric testing of tools like the PHQ-9. Their review highlighted the importance of reliable and validated screening tools for effective depression management in primary care settings [10]. Moreover, advances in technology have opened new avenues for depression screening and assessment. Namboodiri Sandhya Parameswaran and D. Venkataraman introduced a computer vision-based image processing system in their 2019 study, aimed at detecting depression among students. Their system utilized facial features displaying positive and negative emotions to predict depression from video data, achieving an accuracy of 64.38% [11]. In 2022, Rita Francese and Pasquale Attanasio introduced a mobile application that integrates images and speech analysis for depression screening, utilizing the Beck Depression Inventory-II (BDI-II) questionnaire. The application aimed to show the correlation between patients' emotional scores and BDI-II questionnaire scores, demonstrating its potential usefulness in clinical practice [12]. Similarly, Jiao Zhang et al. (2023) introduced a depression screening method analyzing patients' audio and image data using a hybrid neural network with electroencephalography (EEG) signals. Their method provided accurate and rapid screening results for depression, though it required administration by a healthcare professional rather than self-evaluation [13].

In summary, the literature review highlights the multifaceted nature of depression and the various factors influencing its prevalence among Thai individuals. It also discusses the limitations of traditional depression screening tools like the PHQ-9 and explores innovative approaches, including picture-based assessments and technology-based solutions. The discussion extends to the application of technology in enhancing depression screening and assessment, showcasing the potential of computer vision and mobile applications in this domain.

3. Methodology

The PHQ-9 is a self-report questionnaire consisting of 9 questions to screen and classify the severity of depression symptoms. Patients can choose to minimize or exaggerate their responses, which can impact the effectiveness of the assessment results. This research developed a web application to assess depression. Each question of the PHQ-9 assessment is delivered through a story with images, text, and sound to increase relaxation, along with evaluating facial emotions. While evaluating, the camera on your computer or mobile device performs face detection and facial emotion analysis using face-api.js, which is developed using TensorFlow to analyze facial features and identify emotions from facial structure.

Decision results of the depression screening that the total score from the evaluation, by answering questions with the PHQ-9, corresponded to the time value of processing the emotional state that occurred throughout the duration of the assessment. Screening for depression was divided into 4 levels as follows:

1. None: no symptoms of depression: PHQ-9 total score $\leq$ 7, with the highest expression of emotion being happiness or surprise.
2. Mild depression symptoms: PHQ-9 total score = 7-12, with the maximum time of expression of emotion being disgust.
3. Moderate depression symptoms: PHQ-9 total score = 13-18, with the highest time value of expression of emotion being sadness.
4. Severe depression symptoms: PHQ-9 total score > 18, with the highest time values of emotion expression being sadness and fear.

The working design of depression screening with PHQ-9 combined with facial emotion analysis is shown in Figure 1.

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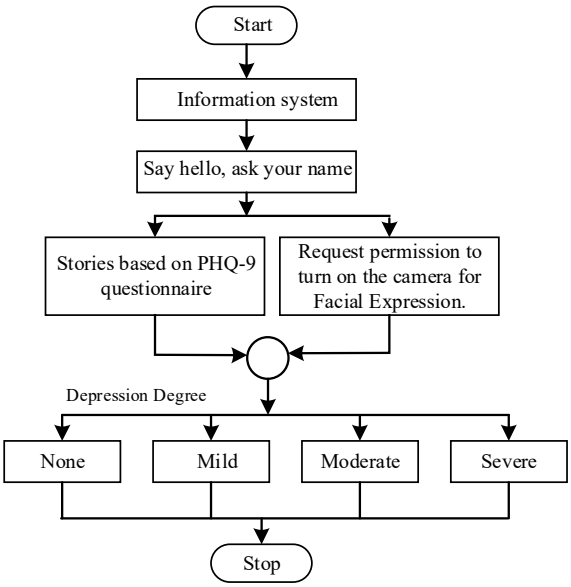

Figure 1 Flowchart of depression screening with PHQ-9 combined with facial emotion analysis.

From Fig. 1 A flowchart illustrates the operation of the web application designed for screening patients with depression. The process begins from the home page and progresses to the website introduction page, where users are prompted to provide a name to be used as a nick name during the assessment. Subsequently, users navigate to the permission page to grant camera access for facial emotion processing and select some choices following the PHQ-9 story. After completing the test, the data is processed, and a summary page is displayed.

The Sampling group for testing web application for depression screening with PHQ-9 combined with facial emotion analysis consists of individuals who meet the following selection criteria: they are aged 15-29 years, as this age group has the highest number of suicides among individuals with depression, and they can provide information by themselves.

Evaluating the effectiveness of depression screening with PHQ-9 combined with facial emotion analysis is based on the testers' average satisfaction scores with using the system, divided into 5 levels as follows:

- Scores between 4.50 and 5.00: highest level.
- Scores between 3.50 and 4.49: high level.
- Scores between 2.50 and 3.49: moderate level.
- Scores between 1.50 and 2.49: low level.
- Scores between 1.00 and 1.49: lowest.

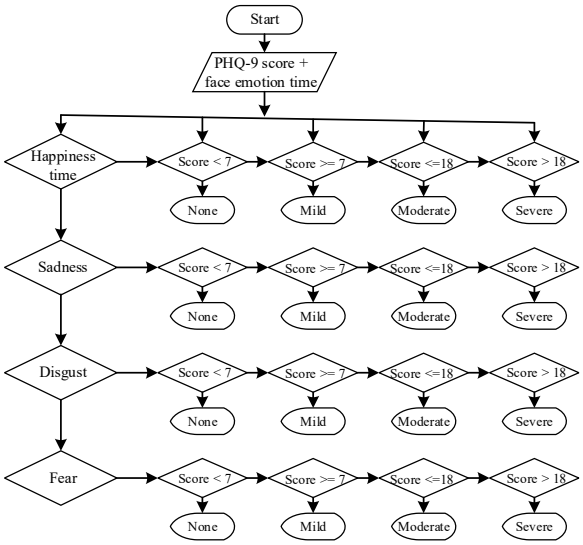

Figure 2 Workflow result assessment of depression screening with PHQ-9 score combined with facial emotion.

3. Results

Results from the study and design of the work procedures mentioned above, focusing on the development of a web application for Depression screening with PHQ-9 combined with facial emotion analysis and test results, are presented in this topic.

(a) (b) (c)

Figure 3 web application screen for Depression screening with PHQ-9 combined with facial emotion analysis.

From Fig. 3(a) The first screen displays information and terms related to using the depression screening with PHQ-9 combined with facial emotion analysis, 3(b) This is the starting page for answering the PHQ-9 assessment questions, 3(c) The sample page for the PHQ-9 questions is shown, which includes images, text, sound, and a button to proceed to the next page.

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Table 1: Results summary of the time (in seconds) of emotions detected.

Q. no.	Neutral (s)	Sad (s)	Happy (s)	Surprised (s)	Disgusted (s)	Fearful (s)	Angry (s)	total
1	6	0	1	1	0	0	0	Neutral
2	9	2	0	1	2	0	0	Neutral
3	7	0	0	0	0	0	0	Neutral
4	1	0	0	0	8	0	0	Disgusted
5	4	0	0	0	1	10	0	Fearful
6	7	0	0	0	0	0	0	Neutral
7	8	0	0	0	2	0	0	Neutral
8	10	0	0	0	2	1	0	Neutral
9	9	0	0	0	1	2	0	Neutral
total	61	2	1	2	16	13	0	

(s: second) is time values indicating the duration of emotions detected from faces while answering the questions.

From table 1 Data on processing time (in seconds) of emotions detected from face detection while answering the 9 questions of the PHQ-9 assessment. Column 1 represents the questions of the PHQ-9, while columns 2-8 display the time values indicating the duration of emotions detected from faces while answering the questions. The emotions include neutral, sad, happy, surprised, disgusted, fearful, and angry. The rightmost column shows the total occurrence of each emotion for every question, determined by the highest time value. The last row displays the total time values for each emotion that occurred while answering all the questions.

4. Conclusion

Depression screening with PHQ-9 combined with facial emotion analysis was developed to effectively increase screening efficiency. Individuals wishing to self-assess now have much easier access to assessments, as it is designed in the form of a web application. Through actual testing, it was found to help create a feeling of relaxation while completing the PHQ-9 standard assessment, thanks to the presentation of the questions through a story with accompanying images, text, and sound. The results of the satisfaction evaluation regarding the overall content of use were found to be at a high level (average 4.10 and standard division 0.44).

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